Hospital Management System - Test Plan (Executed)

90 Pass, 10 Fail. Fails: UT-10, UT-13, IT-10, UT-27, IT-20, UT-31, IT-24, ST-23, ST-24, ST-28.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	Patient Registration & Login	Validate registration with all required patient details.	System is accessible and registration page is available.	1. Open the system. 2. Select Register. 3. Enter all required details. 4. Submit registration.	Name: Ananya Rao; Age: 21; Sex: Female; Phone: 9876543210; Email: ananya.rao@example.com	Entered details are accepted and the registration process proceeds to OTP verification.	Registered through the UI with name, phone, email and password. Account was created (HTTP 201) and the app moved to the 'Verify your phone' OTP screen. The form has no Age or Sex fields, so those test-data values could not be entered.	Pass
UT-02	Patient Registration & Login	Reject registration when a required field is missing.	Registration page is available.	1. Select Register. 2. Leave Email blank. 3. Enter the remaining required details. 4. Submit.	Email: blank; other fields valid	The system validates the input and prevents registration until the required field is provided.	Left Email blank: the browser blocked submission and the page stayed on 'Create your account'. The API also rejects it with 400 'Enter a valid name, email, phone and password.'	Pass
UT-03	Patient Registration & Login	Authenticate a patient using valid login credentials.	A patient account has already been created and verified.	1. Open login. 2. Enter valid credentials. 3. Submit.	Registered phone/email and correct password	The system authenticates the patient and provides access.	Logged in with the registered email and correct password after OTP verification; token issued and the booking page opened. Login by phone number is not supported (phone + password returns 401), so email is the login ID.	Pass
UT-04	Patient Registration & Login	Reject login with invalid credentials.	A patient account exists.	1. Open login. 2. Enter an incorrect password. 3. Submit.	Valid username/phone; Password: incorrect123	The system displays an error and asks the patient to enter credentials again.	Wrong password 'incorrect123' was rejected with 'Invalid email or password.' (401). No token issued; user stayed on the login screen.	Pass
IT-01	Patient Registration & Login	Verify registration-to-OTP integration.	Registration page and OTP service are available.	1. Enter valid registration details. 2. Submit registration. 3. Observe the next step.	Valid patient details	After validation, the system sends an OTP to the registered phone number and moves to OTP entry.	After valid registration the app moved to OTP entry. A 6-digit OTP (hashed, 5-minute expiry) is generated but is only printed in the backend console ('Development OTP for ...'); no SMS is actually sent.	Pass
IT-02	Patient Registration & Login	Verify successful OTP verification creates the patient account.	Valid registration data has been submitted and a correct OTP has been received.	1. Enter the received OTP. 2. Submit verification.	Correct OTP for the registered phone number	The OTP is verified and the patient account is created.	Correct OTP returned 'Phone number verified successfully.' (200). The account (created unverified at registration) became verified and could then log in.	Pass
IT-03	Patient Registration & Login	Verify created account can be used for login.	Registration and OTP verification completed successfully.	1. Open login. 2. Enter the newly created account credentials. 3. Submit.	Newly registered patient credentials	The system authenticates the new account and provides access.	The newly registered and verified account logged in successfully and opened the patient page.	Pass
ST-01	Patient Registration & Login	Complete the end-to-end patient registration and login flow.	System is accessible; OTP delivery is available.	1. Register with valid details. 2. Receive and enter OTP. 3. Complete account creation. 4. Log in with valid credentials.	Valid patient details and received OTP	Registration, OTP verification, account creation, and login complete successfully.	Register -> OTP (read from backend console) -> verified -> login completed end-to-end in the UI. OTP delivery is simulated via the console, not a real SMS.	Pass
ST-02	Patient Registration & Login	Verify invalid login is handled without granting access.	A registered patient account exists.	1. Open login. 2. Enter invalid credentials. 3. Submit. 4. Observe access state.	Incorrect password	An error is displayed and protected patient access is not granted.	Invalid password showed 'Invalid email or password.'; no token was issued and no patient page was displayed.	Pass
ST-03	Patient Registration & Login	Verify patient can retry login after an invalid attempt.	A registered patient account exists.	1. Enter invalid credentials and submit. 2. Correct the credentials. 3. Submit again.	First attempt: incorrect password; second attempt: correct credentials	The invalid attempt is rejected and the subsequent valid attempt grants access.	First attempt with wrong password was rejected; the second attempt with the correct password logged in (UI).	Pass
UT-05	OTP Verification	Generate an OTP for a registered phone number.	A valid registered phone number is available.	1. Provide the registered phone number. 2. Request verification.	Phone: 9876543210	The system generates and sends an OTP to the registered phone number.	A 6-digit OTP is generated at registration for 9876543210 and printed in the backend console (simulated delivery, no SMS gateway). There is no separate 'request OTP' or resend action; the user must register again for a new OTP.	Pass
UT-06	OTP Verification	Accept the correct OTP.	An OTP has been generated and received.	1. Enter the received OTP. 2. Submit.	Correct received OTP	The OTP is validated and the phone number is verified.	Correct OTP accepted: 'Phone number verified successfully.' (200); phone marked verified.	Pass
UT-07	OTP Verification	Reject an incorrect OTP.	An OTP has been generated.	1. Enter an incorrect OTP. 2. Submit.	OTP: 000000	The system rejects the OTP and asks the patient to enter the correct OTP.	OTP 000000 rejected: 'Incorrect OTP. 4 attempts remaining.' (400). Phone stayed unverified and login stayed blocked.	Pass
UT-08	OTP Verification	Reject an empty OTP submission.	OTP entry screen is displayed.	1. Leave the OTP field empty. 2. Submit verification.	OTP: blank	The system does not verify the phone number and prompts for a valid OTP.	Empty OTP: the UI blocks submission (required, 6-digit pattern) and the API returns 400 'Enter a valid phone number and six-digit OTP.' Phone not verified.	Pass
IT-04	OTP Verification	Verify phone number is passed correctly from registration to OTP generation.	Patient registration has been submitted with a valid phone number.	1. Register with a valid phone number. 2. Trigger OTP verification.	Phone: 9876543210	The registered phone number is used for OTP generation and delivery.	The phone entered at registration (9876543210) was the number the OTP was generated for (console: 'Development OTP for 9876543210').	Pass
IT-05	OTP Verification	Verify generated OTP is accepted by the verification component.	An OTP has been generated for the patient.	1. Retrieve the received OTP. 2. Enter it. 3. Submit.	Received OTP	The verification component validates the generated OTP successfully.	The generated OTP (taken from the console) was accepted by the verify-otp endpoint (200).	Pass
IT-06	OTP Verification	Verify successful verification permits continuation to login/registration.	Correct OTP is available.	1. Enter correct OTP. 2. Submit. 3. Continue to the next step.	Correct OTP	The phone number is marked verified and the patient is allowed to continue.	After the correct OTP the phone was marked verified; login, previously blocked with 403 'Please verify your phone number before logging in.', then succeeded.	Pass
ST-04	OTP Verification	Complete the end-to-end OTP verification flow.	System and OTP delivery are available.	1. Provide registered phone number. 2. Request OTP. 3. Enter received OTP. 4. Submit.	Valid registered phone number and correct OTP	OTP is sent, validated, and the phone number is verified.	Register -> OTP generated -> correct OTP entered -> phone verified. OTP delivery is simulated through the backend console.	Pass
ST-05	OTP Verification	Verify incorrect OTP does not allow continuation.	An OTP has been generated.	1. Enter an incorrect OTP. 2. Submit. 3. Observe the next-step access.	Incorrect OTP: 123456	Verification is rejected and continuation is blocked until a correct OTP is entered.	Incorrect OTP was rejected (400, '3 attempts remaining') and login stayed blocked (403) until the correct OTP was verified.	Pass
ST-06	OTP Verification	Verify retry with the correct OTP succeeds after an incorrect attempt.	An OTP has been generated and received.	1. Enter incorrect OTP and submit. 2. Enter the correct OTP. 3. Submit again.	First OTP: incorrect; second OTP: received correct OTP	The first attempt is rejected; the correct OTP then verifies the phone number.	Wrong OTP was rejected; the correct OTP submitted afterwards verified the phone number (200).	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-09	View Medical History	Display stored medical history for an authorized patient.	Patient account is logged in and has stored medical history.	1. Open medical history. 2. Observe the displayed information.	Existing patient medical history	The system retrieves and displays the available medical information and reports.	'My Medical History' shows diagnoses/treatment notes (including doctor-entered diagnosis and 'Prescription issued' entry) and a 'Medical Reports' list with the report requests and files.	Pass
UT-10	View Medical History	Handle a patient record with no stored medical history.	Authorized user is logged in; selected patient has no stored history.	1. Open medical history.	Patient record with no history	The system displays the available information state without exposing unrelated records.	A patient with no history of their own is shown two sample records (Seasonal Allergic Rhinitis, Routine Health Check-up) plus a demo report. The backend returns records with patientId 'demo-patient' to every patient, so unrelated sample records are exposed.	Fail
UT-11	View Medical History	Allow an authorized doctor to view patient history.	Doctor is logged in and authorized for the patient record.	1. Select the patient's medical history. 2. Observe the result.	Authorized doctor account; existing patient record	The patient's stored medical history and reports are displayed.	Doctor dashboard shows 'Medical history' and 'Reports and scans' for an appointment assigned to that doctor (diagnosis, prescription entry, report requests).	Pass
UT-12	View Medical History	Deny access to medical history for an unauthorized user.	User is logged in but is not authorized to access the selected patient record.	1. Attempt to open the patient's medical history.	Unauthorized user; protected patient record	Access is denied and the patient's medical information is not displayed.	No token -> 401; doctor token on patient endpoint -> 403; a second patient did not receive the first patient's diagnosis; a doctor cannot open another doctor's appointment (404). There is no endpoint to request another patient's record by ID.	Pass
IT-07	View Medical History	Verify login state is integrated with medical-history access.	A valid user can log in and has authorization.	1. Log in. 2. Open medical history. 3. Observe the retrieved record.	Valid credentials; authorized patient	The authenticated user's authorization is recognized and the correct medical history is retrieved.	The login token is used to retrieve history; only the logged-in patient's records (plus the shared sample records, see UT-10) were returned.	Pass
IT-08	View Medical History	Verify stored reports are retrieved with the patient's history.	Patient record contains medical history and reports.	1. Open the patient's medical history. 2. Review the returned information.	Patient record with history and reports	Available medical history and reports are retrieved and shown together.	History and reports are returned by one call (/patient/medical-records) and displayed together: diagnosis, CBC/X-ray requests, uploaded lab file and demo report.	Pass
IT-09	View Medical History	Verify authorization check occurs before record retrieval.	User is logged in but unauthorized for the target patient.	1. Request the target medical history. 2. Observe whether protected data is returned.	Unauthorized user; target patient record	Authorization fails and protected medical information is not returned.	Authorization is checked before data is returned: no token 401, wrong role 403, other patient's data never returned.	Pass
ST-07	View Medical History	View medical history end-to-end as a patient.	Patient is registered, logged in, and has stored records.	1. Log in. 2. Select medical history. 3. Review displayed information.	Valid patient account with stored history	The patient's available medical history and reports are displayed.	Patient login -> 'My Medical History' showed stored diagnosis, notes and reports (sample records also shown, see UT-10).	Pass
ST-08	View Medical History	View medical history end-to-end as an authorized doctor.	Doctor is logged in and authorized to access a patient.	1. Log in as doctor. 2. Select patient history. 3. Review displayed information.	Authorized doctor account; patient record	The authorized doctor can view the patient's stored medical information.	Doctor login -> selected an assigned appointment -> patient's stored history and reports were displayed.	Pass
ST-09	View Medical History	Verify unauthorized access is blocked end-to-end.	A protected patient record exists and an unauthorized account is available.	1. Log in as unauthorized user. 2. Attempt to access the patient's history.	Unauthorized account; protected patient record	The system denies access and does not display the patient's medical information.	Unauthorized access blocked (401/403/404 as in UT-12); no medical data returned. Only one doctor login (DOC001) exists, so 'unauthorized doctor' was checked via the assignment rule.	Pass
UT-13	Doctor Availability & Appointment Booking	Display doctors for a selected medical field.	Patient is logged in.	1. Select a medical field. 2. Observe the doctor list.	Medical field: Cardiology	The system displays doctors available for the selected medical field.	No 'medical field' selection exists. The booking form has one 'Choose a doctor' dropdown listing all 4 doctors labelled with specialty (Cardiology, Dermatology, General Medicine, Orthopedics). The API ignores a specialty filter and returns all 4.	Fail
UT-14	Doctor Availability & Appointment Booking	Display available appointment schedules for a selected doctor.	Patient is logged in and a doctor is available.	1. Select a doctor. 2. View appointment schedules.	Doctor: Dr. A; field: Cardiology	Available appointment schedules are displayed.	Selecting a doctor and date displays the available time slots (09:00-16:00, 24-hour format).	Pass
UT-15	Doctor Availability & Appointment Booking	Book an available appointment slot.	Patient is logged in and the selected slot is available.	1. Select a medical field. 2. Select a doctor. 3. Select an available slot. 4. Confirm booking.	Doctor: Dr. A; Slot: 10:00 AM, 15-Sep-2026	The appointment is confirmed and stored; confirmation is displayed.	Booked Dr. Ananya Rao (Cardiology), 10:00 on 19-Nov-2026: 'Appointment booked successfully.' and it appears under Upcoming appointments. Test-data date 15-Sep-2026 is in the past and is rejected by design ('Appointments cannot be booked in the past'), so a future date was used.	Pass
UT-16	Doctor Availability & Appointment Booking	Prevent booking when no appointment slot is available.	Patient is logged in and selected doctor has no available slots.	1. Select medical field. 2. Select doctor. 3. View schedules.	Doctor with all slots occupied	The system informs the patient that no slot is available and displays other available schedules/doctors if present.	With all 7 slots of a doctor/day taken, the API returns no slots and booking is rejected (409 'This time slot is unavailable'). UI shows 'No slots available for this date. Try another date.' (per code); other doctors/dates are not listed automatically.	Pass
IT-10	Doctor Availability & Appointment Booking	Verify medical-field selection retrieves the correct doctors.	Patient is logged in; doctor data exists for multiple fields.	1. Select a medical field. 2. Observe returned doctors.	Field: Cardiology	Only relevant available doctors are displayed for the selected field.	No field-based retrieval: /api/doctors returns all doctors regardless of specialty and the UI has no field selector, so 'only relevant doctors for the selected field' cannot be shown.	Fail
IT-11	Doctor Availability & Appointment Booking	Verify doctor selection retrieves current appointment schedules.	A doctor has configured appointment schedules.	1. Select the doctor. 2. View available schedules.	Selected doctor with available and occupied slots	The system displays the doctor's current available appointment schedules.	Selecting a doctor and date loads the current slots; the booked 10:00 slot disappeared from the list after booking.	Pass
IT-12	Doctor Availability & Appointment Booking	Verify booking stores the selected slot and returns confirmation.	Patient is logged in; selected slot is available.	1. Select doctor and slot. 2. Confirm booking. 3. Reopen booked appointments.	Doctor: Dr. A; Slot: 10:00 AM	The selected appointment is stored and a confirmation is displayed.	Booking was stored (status Booked), confirmation displayed, and the appointment reappears in 'Upcoming appointments' / GET /appointments/me.	Pass
ST-10	Doctor Availability & Appointment Booking	Book an appointment through the complete patient flow.	Patient account exists; doctor and appointment data are available.	1. Log in. 2. Select medical field. 3. Select doctor. 4. Select available slot. 5. Confirm booking.	Cardiology; Dr. A; available slot	The appointment is successfully booked and confirmation is displayed.	Login -> choose doctor (specialty shown in the label) -> date -> slot -> Confirm: booked and confirmation shown. There is no separate 'select medical field' step.	Pass
ST-11	Doctor Availability & Appointment Booking	Verify unavailable-slot handling through the complete flow.	Patient is logged in; selected doctor has no free slots.	1. Select medical field. 2. Select doctor. 3. View schedules.	Doctor with no available slots	The system informs the patient that no appointment slot is available and offers other schedules/doctors where available.	Fully booked doctor/day: no slots offered and further booking rejected (409). Message suggests another date; alternative doctors are not suggested automatically.	Pass
ST-12	Doctor Availability & Appointment Booking	Verify booked appointment persists after returning to the appointment section.	A patient has successfully booked an appointment.	1. Complete a booking. 2. Navigate away. 3. Reopen booked appointments.	Previously booked doctor and slot	The booked appointment remains stored and is shown in the patient's appointments.	After logging out and back in, the booked appointment was still listed. Data is held in memory, so it is lost if the backend restarts.	Pass
UT-17	Appointment Cancellation	Display a patient's booked appointments.	Patient is logged in and has at least one booked appointment.	1. Open booked appointments. 2. Observe the list.	Patient with one or more appointments	The patient's booked appointments are displayed.	'Upcoming appointments' lists the patient's booked appointments (doctor, specialty, date/time, status); cancelled ones move to 'Past appointments'.	Pass

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UT-18	Appointment Cancellation	Display details of the appointment selected for cancellation.	Patient has a booked appointment.	1. Open booked appointments. 2. Select an appointment to cancel.	Appointment ID: APT001	The selected appointment details are displayed.	Clicking 'Cancel appointment' shows a confirmation dialog with the doctor, date and time of the selected appointment. There is no separate details page.	Pass
UT-19	Appointment Cancellation	Cancel a booked appointment after confirmation.	Patient has a booked appointment.	1. Select appointment. 2. Review details. 3. Confirm cancellation.	Appointment ID: APT001; Confirm: Yes	The appointment status is updated to cancelled.	Confirmed (OK): 'Appointment cancelled successfully.'; status changed to cancelled and the appointment moved to Past appointments.	Pass
UT-20	Appointment Cancellation	Keep appointment active when cancellation is not confirmed.	Patient has a booked appointment.	1. Select appointment. 2. Review details. 3. Do not confirm cancellation.	Appointment ID: APT001; Confirm: No	The appointment remains active.	Dialog dismissed: the appointment stayed 'Booked' in the Upcoming list and no cancel request was made.	Pass
IT-13	Appointment Cancellation	Verify cancellation updates the appointment record.	A booked appointment exists.	1. Select and confirm cancellation. 2. Reopen the appointment record.	Appointment ID: APT001	The stored appointment status is changed to cancelled.	Stored status changed from Booked to cancelled; cancelling the same appointment again returns 409 'This appointment can no longer be cancelled.'	Pass
IT-14	Appointment Cancellation	Verify a cancelled appointment's slot becomes available again.	A booked appointment occupies a slot.	1. Cancel the appointment. 2. Check the doctor's available schedules.	Appointment slot: 10:00 AM	The cancelled slot becomes available again.	After cancelling the 11:00 appointment, 11:00 reappeared in the doctor's slot list.	Pass
IT-15	Appointment Cancellation	Verify the cancellation status is reflected in the patient's appointment list.	Patient has a booked appointment.	1. Cancel the appointment. 2. Return to booked appointments.	Appointment ID: APT001	The appointment is shown with cancelled status.	The cancelled appointment is shown with 'Cancelled' status under Past appointments.	Pass
ST-13	Appointment Cancellation	Cancel an appointment through the complete patient flow.	Patient is logged in and has a booked appointment.	1. Log in. 2. Open booked appointments. 3. Select appointment. 4. Confirm cancellation.	Appointment ID: APT001	The appointment is cancelled successfully and its status is updated.	Login -> Cancel appointment -> confirm: cancelled successfully and status updated.	Pass
ST-14	Appointment Cancellation	Verify non-confirmed cancellation leaves the appointment unchanged.	Patient has an active appointment.	1. Open the appointment. 2. Select cancel. 3. Do not confirm. 4. Recheck appointment.	Appointment ID: APT001; Confirm: No	The appointment remains active.	Dismissing the confirmation left the appointment active; it was unchanged on recheck.	Pass
ST-15	Appointment Cancellation	Verify cancelled slot can be booked again.	An appointment occupies a slot and the patient can cancel it.	1. Cancel the appointment. 2. Open doctor availability. 3. Check the previously occupied slot.	Cancelled appointment slot: 10:00 AM	The previously occupied slot is available for booking again.	The cancelled 11:00 slot was offered again and was booked again successfully (201).	Pass
UT-21	Medical Diagnosis & Reports	Display available patient medical history to the doctor.	Doctor is logged in and has access to a patient appointment/record.	1. Open the patient's appointment/record. 2. Review medical history.	Patient record with existing history	The patient's available medical history is displayed.	Doctor dashboard shows 'Medical history' for the selected assigned appointment (empty list when the patient has no stored history).	Pass
UT-22	Medical Diagnosis & Reports	Enter and save an initial diagnosis.	Doctor is viewing an authorized patient record.	1. Enter initial diagnosis. 2. Save the record.	Diagnosis: Suspected viral infection	The initial diagnosis is accepted and saved to the patient's record.	Diagnosis 'Suspected viral infection' saved: 'Diagnosis saved.' (201) and it appears in the patient's records. A blank diagnosis is rejected (400).	Pass
UT-23	Medical Diagnosis & Reports	Specify required reports or scans.	Doctor is viewing the patient's record.	1. Select reports/scans. 2. Specify the required items. 3. Save.	Required report: CBC; Scan: X-ray	Required reports/scans are recorded with the patient record.	Report/scan requests 'CBC' and 'X-ray' were recorded (status Requested) against the patient; an optional PDF/JPG/PNG attachment also works. Reports and scans are entered as free text; there is no separate report/scan type.	Pass
UT-24	Medical Diagnosis & Reports	Proceed with diagnosis when some patient information is unavailable.	Doctor is logged in; some patient information is unavailable.	1. Open the patient record. 2. Review available information. 3. Enter diagnosis. 4. Save.	Available information only; Diagnosis: Initial assessment	The doctor can proceed using available information and enter the diagnosis.	Diagnosis 'Initial assessment' saved without treatment notes and for a patient with no prior history (201).	Pass
IT-16	Medical Diagnosis & Reports	Verify patient history is available to diagnosis workflow.	Doctor is authorized and patient record contains history.	1. Open patient appointment. 2. Review retrieved history. 3. Enter diagnosis.	Patient history; diagnosis text	Retrieved patient information is available before diagnosis entry.	The doctor's appointment data includes the patient's history and reports, so they are available before the diagnosis is entered.	Pass
IT-17	Medical Diagnosis & Reports	Verify diagnosis and required reports are stored in the same patient record.	Doctor has access to patient record.	1. Enter diagnosis. 2. Add required reports/scans. 3. Save. 4. Reopen record.	Diagnosis: Viral infection; Report: CBC; Scan: X-ray	Diagnosis and required reports/scans are saved and associated with the correct patient record.	Diagnosis and the CBC/X-ray requests were stored against the same patient and appointment, and both are visible after reopening in the doctor and patient views.	Pass
IT-18	Medical Diagnosis & Reports	Verify saved diagnosis can be retrieved after the consultation workflow.	A diagnosis has been saved for the patient.	1. Save diagnosis. 2. Close/reopen patient record. 3. Review diagnosis.	Saved diagnosis: Initial assessment	The saved diagnosis is retrievable from the patient's record.	The saved diagnosis is still shown after reloading and after a new login (doctor and patient views).	Pass
ST-16	Medical Diagnosis & Reports	Complete the doctor diagnosis and report workflow.	Doctor is logged in and an appointment/patient record exists.	1. Open patient appointment. 2. Review history. 3. Enter diagnosis. 4. Specify reports/scans. 5. Save.	Diagnosis and required report/scan values	Diagnosis and required reports/scans are saved to the patient's record.	Doctor login -> open appointment -> review history -> enter diagnosis -> record report/scan -> saved with confirmation messages.	Pass
ST-17	Medical Diagnosis & Reports	Verify diagnosis can be completed with partial patient information.	Some patient information is unavailable.	1. Open patient record. 2. Review available information. 3. Enter diagnosis. 4. Save.	Partial record; Diagnosis: Initial assessment	The doctor can proceed with available information and save the diagnosis.	Diagnosis saved using only partial information (no treatment notes, no prior history).	Pass
ST-18	Medical Diagnosis & Reports	Verify stored diagnosis and reports are visible in the patient record after completion.	Doctor has completed and saved diagnosis/report details.	1. Save diagnosis and report requirements. 2. Reopen patient record. 3. Review stored information.	Previously saved diagnosis and report/scan details	The saved diagnosis and required reports/scans remain associated with the patient record.	After saving and re-login, the diagnosis and report requests remain in the patient record. Data is held in memory and is lost on backend restart.	Pass
UT-25	Prescription Management	Allow doctor to enter final prescription and medication details.	Doctor is logged in and has opened the patient's medical record.	1. Enter final prescription. 2. Enter medication details. 3. Save.	Prescription: Amoxicillin 500 mg; Frequency: 2 times/day	Prescription details are accepted for validation and storage.	Prescription saved ('Prescription saved successfully.', 201) with medicine, dosage and frequency. Amoxicillin is not in the demo catalogue (Paracetamol, Cetirizine, Vitamin C, Antacid), so Paracetamol 500 mg, 3/day was used.	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-26	Prescription Management	Reject incomplete prescription details.	Doctor is logged in and prescription form is open.	1. Leave required medication information incomplete. 2. Attempt to save.	Medication name: blank; dosage: 500 mg	The system asks the doctor to provide the required information before saving.	Blank medicine (or missing dosage/instructions) is rejected: the UI form is blocked as invalid and the API returns 400 'Choose distinct medicines and enter valid dosage, frequency, and instructions.'	Pass
UT-27	Prescription Management	Display a stored prescription to the patient.	A valid prescription has already been stored for the patient.	1. Log in as patient. 2. Open prescription section.	Stored prescription for patient	The stored prescription is displayed.	Patient's 'My Prescriptions' section shows only a generic 'Prescription' title, the date and a Download button; medicine, dosage and instructions are missing because the component reads top-level fields while the API returns them in items[]. Full details appear only under 'Medicine Purchase > My prescriptions'.	Fail
UT-28	Prescription Management	Validate and store complete prescription details.	Doctor is logged in and patient record is available.	1. Enter complete prescription details. 2. Save. 3. Reopen record.	Medicine: Paracetamol 500 mg; Frequency: 3/day; Duration: 3 days	The prescription passes validation, is stored, and remains available in the record.	Complete prescription (Paracetamol 500 mg; 3/day; 'After food for 3 days') was validated, stored and is still present after reopening. There is no separate 'Duration' field, so duration goes in Instructions.	Pass
IT-19	Prescription Management	Verify prescription is associated with the correct patient record.	Doctor has access to a specific patient record.	1. Open patient record. 2. Enter prescription. 3. Save. 4. Reopen the record.	Patient: P001; Prescription: Paracetamol 500 mg	The prescription is stored against the selected patient and not another record.	Prescription was stored against the appointment's patient; a second patient's prescription list returned 0 items.	Pass
IT-20	Prescription Management	Verify stored prescription is retrieved by the patient-facing prescription section.	A prescription has been saved successfully.	1. Save prescription as doctor. 2. Log in as the patient. 3. Open prescriptions.	Stored prescription	The patient-facing section retrieves and displays the saved prescription.	The saved prescription is retrieved correctly from the API, but the patient-facing 'My Prescriptions' section does not display its medicine details (see UT-27).	Fail
IT-21	Prescription Management	Verify incomplete prescription is not stored.	Doctor is logged in and form is open.	1. Enter incomplete details. 2. Attempt to save. 3. Reopen the patient record.	Missing medication name	The incomplete prescription is rejected and is not saved as a valid prescription.	Incomplete prescription was rejected (400) and not stored; the patient's prescription count did not change.	Pass
ST-19	Prescription Management	Complete prescription creation and patient viewing end-to-end.	Doctor and patient accounts exist; patient record is available.	1. Log in as doctor. 2. Open patient record. 3. Enter and save prescription. 4. Log in as patient. 5. Open prescription section.	Complete prescription details	Doctor can save the prescription and the patient can view the stored prescription.	Doctor saved the prescription; the patient can view it (full details under Medicine Purchase > My prescriptions) and download it. The 'My Prescriptions' card itself is incomplete, see UT-27.	Pass
ST-20	Prescription Management	Verify incomplete prescription handling end-to-end.	Doctor is logged in and patient record is available.	1. Enter incomplete prescription. 2. Attempt to save. 3. Correct missing details. 4. Save again.	First: missing medication name; second: complete details	The incomplete attempt is rejected; the corrected prescription is successfully stored.	Incomplete prescription was rejected, then a corrected one with all required fields was saved successfully.	Pass
ST-21	Prescription Management	Verify the stored prescription remains available after a new login session.	A prescription has been successfully stored.	1. Log out. 2. Log back in as patient. 3. Open prescription section.	Previously stored prescription	The stored prescription is still displayed to the patient.	After logging out and back in as the patient, the stored prescription was still returned and displayed. Data is held in memory and is lost on backend restart.	Pass
UT-29	Online Payment & Medicine Purchase	Calculate payment amount for selected prescribed medicines.	Patient has prescribed medicines available.	1. View prescribed medicines. 2. Select required medicines. 3. Observe calculated amount.	Medicine A: quantity 1; Medicine B: quantity 2	The system calculates and displays the payment amount for the selected medicines.	Server-side total for Paracetamol (Rs 30) x1 + Cetirizine (Rs 45) x2 = Rs 120 (API); the UI shows the running total for the selected items (Rs 30.00 for one item).	Pass
UT-30	Online Payment & Medicine Purchase	Accept valid online payment details.	Patient has selected medicines and reached payment.	1. Select online payment. 2. Enter valid payment details. 3. Submit payment.	Valid test payment details	The payment is processed securely and a successful payment confirmation is displayed when the payment succeeds.	Checkout -> 'Pay Now (Demo)' -> 'Demo payment successful. No real money was charged. Reference: DEMO-...'. Payment is simulated; there is no card/payment-detail form, so the 'enter payment details' step does not exist.	Pass
UT-31	Online Payment & Medicine Purchase	Handle failed payment.	Patient has selected medicines and reached payment.	1. Enter payment details that cause a payment failure. 2. Submit.	Payment data configured to fail in the test environment	The system displays payment failure and allows the patient to retry.	No failure path exists: the simulated payment always succeeds (even when failure-style data was sent), and there is no payment-failure message or retry prompt.	Fail
UT-32	Online Payment & Medicine Purchase	Record a successful medicine purchase.	A payment has completed successfully.	1. Complete successful payment. 2. Review purchase status/history.	Successful medicine payment	The medicine purchase is recorded.	After payment the order shows 'Paid (Demo)' with a transaction reference in 'My medicine orders', and the bill shows Paid.	Pass
IT-22	Online Payment & Medicine Purchase	Verify selected medicines feed into payment calculation.	Prescribed medicines and prices are available.	1. Select one or more prescribed medicines. 2. Proceed to payment. 3. Compare selected items with calculated amount.	Selected medicines with known test prices	The payment amount corresponds to the selected medicines.	Order amount equals the sum of selected prescribed items x server prices; a non-prescribed medicine is rejected (403) and quantity 0 is rejected (400).	Pass
IT-23	Online Payment & Medicine Purchase	Verify successful payment triggers purchase recording.	Medicine selection is complete and payment can succeed.	1. Submit valid payment. 2. Wait for success confirmation. 3. Check purchase record.	Valid test payment details	Successful payment is confirmed and the medicine purchase is recorded.	Successful payment returned a confirmation with reference and the order was recorded as 'Paid (Demo)'; paying the same order again returns 409.	Pass
IT-24	Online Payment & Medicine Purchase	Verify payment failure returns the user to a retry-capable state.	Medicine selection is complete.	1. Submit failing payment. 2. Observe failure response. 3. Attempt retry.	Failing payment data, then valid test payment data	Failure is displayed and the patient is allowed to retry payment.	A payment failure cannot be triggered, so no failure or retry state exists; an unpaid order simply stays 'Pending' until paid.	Fail
ST-22	Online Payment & Medicine Purchase	Complete medicine selection, payment, confirmation and purchase recording.	Patient has a valid prescription and prescribed medicines are available.	1. View medicines. 2. Select medicines. 3. Review calculated amount. 4. Select online payment. 5. Enter valid test payment details. 6. Submit.	Prescribed medicines; valid test payment details	Payment succeeds, confirmation is displayed, and the medicine purchase is recorded.	View medicines -> select -> total shown -> Proceed to Checkout -> Pay Now (Demo) -> confirmation -> order recorded as 'Paid (Demo)'.	Pass
ST-23	Online Payment & Medicine Purchase	Verify failed payment does not produce a false successful purchase.	Patient has selected medicines.	1. Submit a failing payment. 2. Observe result. 3. Check purchase record.	Failing payment data	Payment failure is shown and a successful purchase is not recorded.	Cannot be verified: no payment-failure scenario is possible (every payment succeeds), so 'failure shown, no purchase recorded' is not implemented.	Fail
ST-24	Online Payment & Medicine Purchase	Retry a failed payment successfully.	Patient has selected medicines and the first payment attempt failed.	1. Submit failing payment. 2. Retry with valid test payment details. 3. Submit.	First: failing test data; second: valid test payment data	The first attempt fails; the retry succeeds and the purchase is recorded.	Retry-after-failure flow is not implemented because the first payment attempt cannot fail.	Fail

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-33	File Download	Display available medical documents.	Patient is logged in and documents exist.	1. Open medical documents section. 2. Observe available files.	Reports and prescriptions stored for patient	Available reports, prescriptions and other files are displayed.	'Medical Reports' lists each report (name, status, file name) with a 'Download Report' button when a file exists. Prescriptions are listed separately under 'My Prescriptions', not in the same document list.	Pass
UT-34	File Download	Select an available file for download.	Medical documents section contains an available file.	1. Open documents. 2. Select a file.	File: Prescription_P001.pdf	The selected file is identified and ready for download.	Report file Report_P001.pdf (uploaded by the doctor) was selectable via its 'Download Report' button. No Prescription_P001.pdf exists; prescriptions are exported as JSON.	Pass
UT-35	File Download	Download an available medical file.	Patient is logged in and selected file is available.	1. Select the file. 2. Select Download. 3. Check the device download location.	File: Report_P001.pdf	The selected file is downloaded to the patient's device.	'Download Report' saved the file (demo-medical-report.pdf, 1,056 bytes, valid %PDF header). API download of the uploaded Report_P001.pdf returned 200 with the correct filename.	Pass
UT-36	File Download	Handle a selected file that is unavailable.	Patient is logged in; requested file is unavailable.	1. Open documents. 2. Attempt to select/download the unavailable file.	Missing file reference	The system displays an appropriate error message.	With the file removed from server storage the UI showed 'The report file is missing from server storage.' (404); an unknown report ID returns 'Report file not found.'	Pass
IT-25	File Download	Verify medical-document records are connected to the patient's document list.	Patient has stored reports/prescriptions.	1. Log in. 2. Open medical documents. 3. Compare displayed files with stored documents.	Stored report and prescription	Available stored documents are retrieved for the correct patient.	Only the logged-in patient's documents are listed; a second patient saw only their own demo report and gets 403 when requesting the first patient's file.	Pass
IT-26	File Download	Verify selected file is passed to the download function.	An available file is displayed.	1. Select a file. 2. Select Download.	File: Prescription_P001.pdf	The requested file is passed to the download operation and download begins.	The selected report ID was passed to GET /reports/:id/download and the download started with the correct filename and content type.	Pass
IT-27	File Download	Verify unavailable-file error is returned by the download operation.	A file reference exists but the underlying file is unavailable.	1. Select the unavailable file. 2. Select Download.	Missing file reference	The system returns an appropriate error instead of downloading a nonexistent file.	Missing file returns 404 'The report file is missing from server storage.'; no file is returned.	Pass
ST-25	File Download	Download a medical report through the complete patient flow.	Patient is logged in and has an available report.	1. Open medical documents. 2. Select report. 3. Select Download. 4. Verify the downloaded file.	Available report file	The selected report downloads successfully to the patient's device.	Report downloaded through the UI as a valid PDF (1,056 bytes).	Pass
ST-26	File Download	Download a prescription through the complete patient flow.	Patient is logged in and has a stored prescription file.	1. Open medical documents. 2. Select prescription. 3. Select Download. 4. Verify the downloaded file.	Available prescription file	The selected prescription downloads successfully.	'Download Prescription' downloaded prescription-<id>.json (495 bytes) containing the prescription data. It is a JSON export, not a PDF.	Pass
ST-27	File Download	Verify unavailable-file handling through the complete flow.	Patient is logged in and selected file is unavailable.	1. Open medical documents. 2. Select unavailable file. 3. Select Download.	Unavailable file	An appropriate error is displayed and no invalid file is downloaded.	Unavailable file: an error message is displayed and no invalid file is downloaded.	Pass
UT-37	Bed Availability	Retrieve current bed information.	Bed availability section is accessible.	1. Open bed availability. 2. Observe retrieved bed information.	Current bed dataset	The system retrieves current bed information.	Doctor dashboard now has a 'Bed availability' section that retrieves the bed inventory (16 beds in 4 wards) with Total/Available/Occupied counts, a ward filter and a 5-second auto-refresh. GET /api/beds is doctor-only.	Pass
UT-38	Bed Availability	Display available beds as available.	At least one required bed is available.	1. Open bed availability. 2. Observe bed status.	Test data: required bed status = available	The system displays the bed as available.	Available beds (e.g. GW-102) are shown with status 'Available' and a patient-name box plus 'Allocate bed' button.	Pass
UT-39	Bed Availability	Display occupied beds as occupied.	Required beds are occupied.	1. Open bed availability. 2. Observe bed status.	Test data: required bed status = occupied	The system displays the bed as occupied.	Occupied beds (e.g. GW-101) are shown with status 'Occupied', the patient name and a 'Release bed' button.	Pass
UT-40	Bed Availability	Display all required beds as occupied when no beds are available.	No required beds are available.	1. Open bed availability. 2. Observe the availability message.	All required beds = occupied	The system displays that all required beds are currently occupied.	With every bed allocated, the API returned total 16 / available 0 / occupied 16, and the UI renders these counts and 'Occupied' on each bed. There is no dedicated 'all beds occupied' message, and the all-occupied state was checked at API level only.	Pass
IT-28	Bed Availability	Verify bed data retrieval feeds the availability check.	Current bed records exist.	1. Access bed availability. 2. Retrieve current bed information. 3. Check status.	Current bed records	The availability check uses the current retrieved bed information.	Counts are computed from the current bed records (16 total, 10 available, 6 occupied); the ward filter returned only that ward (ICU: 4 beds, 2 available, 2 occupied).	Pass
IT-29	Bed Availability	Verify a bed-status update is reflected in the availability display.	A test bed record can be changed between available and occupied.	1. Set test bed status. 2. Refresh/open availability. 3. Observe status.	Bed B001: available, then occupied	The displayed availability reflects the current stored status.	Allocating GW-102 changed it to Occupied and updated the summary (available 10 -> 9, occupied 6 -> 7); releasing it returned it to Available. The UI shows the update straight after the action. Test beds B001/B002 do not exist; real IDs were used.	Pass
IT-30	Bed Availability	Verify the availability display distinguishes available and occupied states.	Test data contains both available and occupied beds.	1. Open bed availability. 2. Review multiple bed statuses.	B001 available; B002 occupied	Each bed is displayed with the correct availability state.	Available and occupied beds are displayed distinctly in the same list (e.g. ICU-201 Occupied with patient name, ICU-202 Available). Allocating an occupied bed returns 409; releasing an available bed returns 409.	Pass
ST-28	Bed Availability	Check bed availability end-to-end as a patient.	Patient can access the bed availability section; current bed data exists.	1. Log in. 2. Open bed availability. 3. Review status.	Current bed data	The system displays current bed availability as available or occupied.	Patients cannot check bed availability: the bed section exists only on the doctor dashboard, no bed section appears on the patient page, and GET /api/beds with a patient token returns 403 'Doctor access only.'	Fail
ST-29	Bed Availability	Check bed availability end-to-end as hospital staff.	Hospital staff account is available and current bed data exists.	1. Log in as staff. 2. Open bed availability. 3. Review status.	Staff account; current bed data	The system displays current bed availability to authorized hospital staff.	Logged in with the doctor (staff) account and the Bed availability section showed current bed status. Doctor is the only staff login; there is no separate hospital-staff role.	Pass
ST-30	Bed Availability	Verify no-bed-available condition end-to-end.	No required beds are available in the test data.	1. Access bed availability. 2. Observe the displayed status/message.	All required beds occupied	The system displays that all required beds are currently occupied.	When all beds are occupied the dashboard shows Available 0 / Occupied 16 and every bed as Occupied (API-verified). There is no dedicated 'all beds occupied' message.	Pass